

COMSATS University Islamabad, Vehari Campus

Information Communication Technology (CSC101)

Lab 10

Topic: Python – Elementary Programming

E-mail: mariamsherwani434@gmail.com

Objective

The objective of this lab is to understand variables and the basics of Python using examples and practical tasks.

Activity Outcomes

- Basics of programming
- Variables
- Print function
- Eval function

Pre-Lab Reading

Read Chapter 3 from *Python Basics: A Practical Introduction to Python 3 (2021)*.

1. Useful Concepts

1. Print Function

A function is a block of code that takes input and produces output.

Syntax:

```
print(<thing to display>)
```

2. Identifiers

Identifiers are names used for variables, functions, etc.

Rules:

- Can contain letters, digits, underscores (_), dollar sign (\$)
 - Cannot start with a digit
 - Cannot be a reserved keyword
 - Cannot be: true, false, null
-

3. Variables

Variables store values temporarily.

Syntax:

```
variable_name = value
```

4. Statements

A line of code that performs an action is called a statement.

5. Assignment

Assigning a value to a variable is called assignment.

Example:

```
x = 10
```

6. Expressions

Combination of values, variables, and operators.

Example:

```
x = 5 * (3 / 2) + 3 * 2
```

7. Data Types

- **int** → integers
 - **float** → decimal numbers
 - **str** → text
 - **bool** → True/False
-

8. Integer and Float Example

```
num1 = 34  
num2 = 3.21
```

9. Strings

Text enclosed in quotes.

```
x = 'This is a tomato'
```

10. Boolean

```
y = True  
z = False
```

11. Comments

Used to explain code.

Single line:

```
# This is a comment
```

Multi-line:

```
'''  
  
This is a  
multiline comment  
'''
```

12. eval() Function

Evaluates a string as a Python expression.

Syntax:

```
eval(expression)
```

2. Solved Lab Activities

Activity 1: Python program to do some math operations on numbers

(Math Operations):

```
x = 2
y = 3
z = 1
result = 2 + 3 - 1
print(result)
```

Output:

4

Activity 2: Python program to illustrate a string (String Example):

```
x = "This is Python"
y = "This is Java"
print(x)
print(y)
```

Output:

This is Python
This is Java

Activity 3: Write a program that takes three numbers and prints their sum. Every number is given on a separate line (Sum of Three

Numbers):

```
a = int(input())
b = int(input())
c = int(input())
print(a + b + c)
```

Input:

2
3
5

Output:

10

Activity 4: Write a program that reads an integer number and prints its previous and next numbers (Previous and Next Number):

```
a = int(input())
print("The next number for the number", a, "is", a+1)
print("The previous number for the number", a, "is", a-1)
```

Input:

179

Output:

The next number for the number 179 is 180.
The previous number for the number 179 is 178.

Activity 5: Python program to illustrate comments (Comments

Example):

```
# Addition Function  
'''  
Addition Function  
This function takes in two numbers and returns the sum  
@param1 number  
@param2 number  
@returns number  
'''
```

Output:

No output

3. Graded Lab Tasks (Solutions)

Task 1: Write a program that greets the person by printing the word "Hi" and the name of the person.

Greeting Program:

```
name = input("Enter your name: ")  
print("Hi", name)
```

Example Input:

Ali

Output:

Hi Ali

Task 2: Write a program that takes a number and print its square.

Square of a Number:

```
num = int(input("Enter a number: "))  
print("Square is:", num * num)
```

Example Input:

5

Output:

Square is: 25

Task 3: Write a program that reads the length of the base and the height of a right-angled triangle and prints the area. Every number is given on a separate line.

Area of Right-Angled Triangle:

```
base = float(input("Enter base: "))  
height = float(input("Enter height: "))  
area = 0.5 * base * height  
print("Area of triangle:", area)
```

Example Input:

4

6

Output:

Area of triangle: 12.0

End of Lab Lecture